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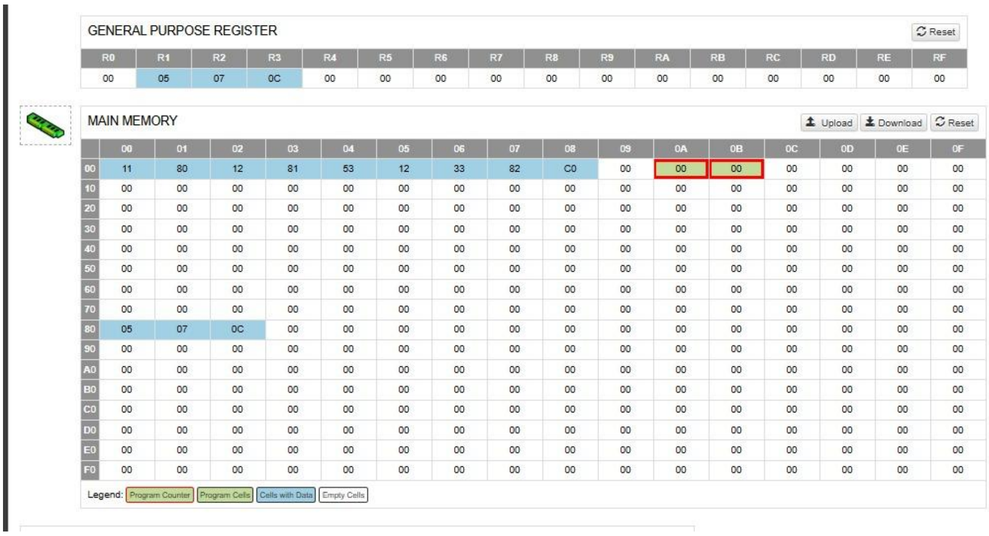
Assignment 3

Task 1



GENERAL PURPOSE REGISTER

R0	R1	R2	R3	R4	R5	R6	R7	R8	R9	RA	RB	RC	RD	RE	RF
00	05	07	0C	00	00	00	00	00	00	00	00	00	00	00	00

MAIN MEMORY

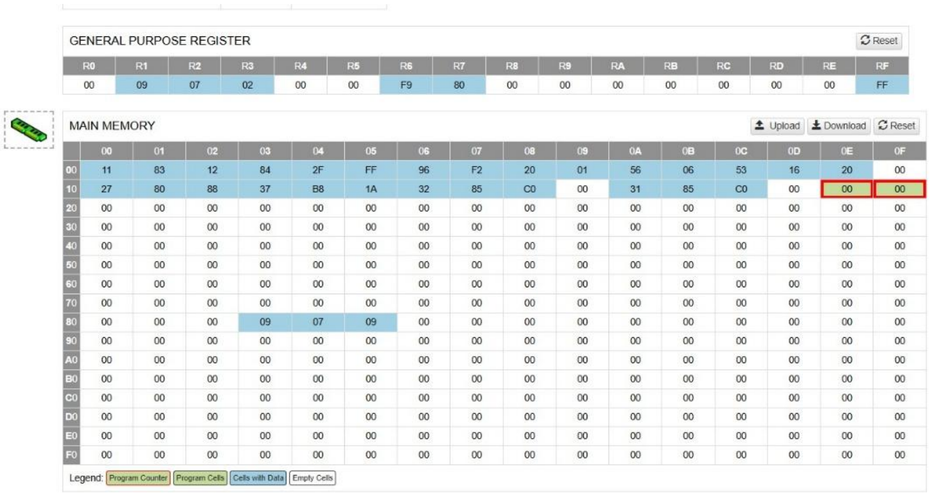
	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
00	11	80	12	81	53	12	33	82	C0	00	00	00	00	00	00	00
10	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
20	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
30	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
40	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
50	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
60	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
70	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
80	05	07	0C	00	00	00	00	00	00	00	00	00	00	00	00	00
90	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
B0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
C0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
D0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
F0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

Legend: Program Counter Program Cells Cells with Data Empty Cells

Explanation for the program:

The program begins with two LOAD commands: moves datas from their memory addresses into registers R1 and R2. Then the ADD instruction performs the addition on the contents of R1 and R2, placing the resulting into R3. Following the calculation, the STORE instruction moves the result from R3 and writes it back to Memory address. Finally, the HALT command is executed, which stops all further program execution without any additional data movement.

Task 2



GENERAL PURPOSE REGISTER

R0	R1	R2	R3	R4	R5	R6	R7	R8	R9	RA	RB	RC	RD	RE	RF
00	09	07	02	00	00	F9	80	00	00	00	00	00	00	00	FF

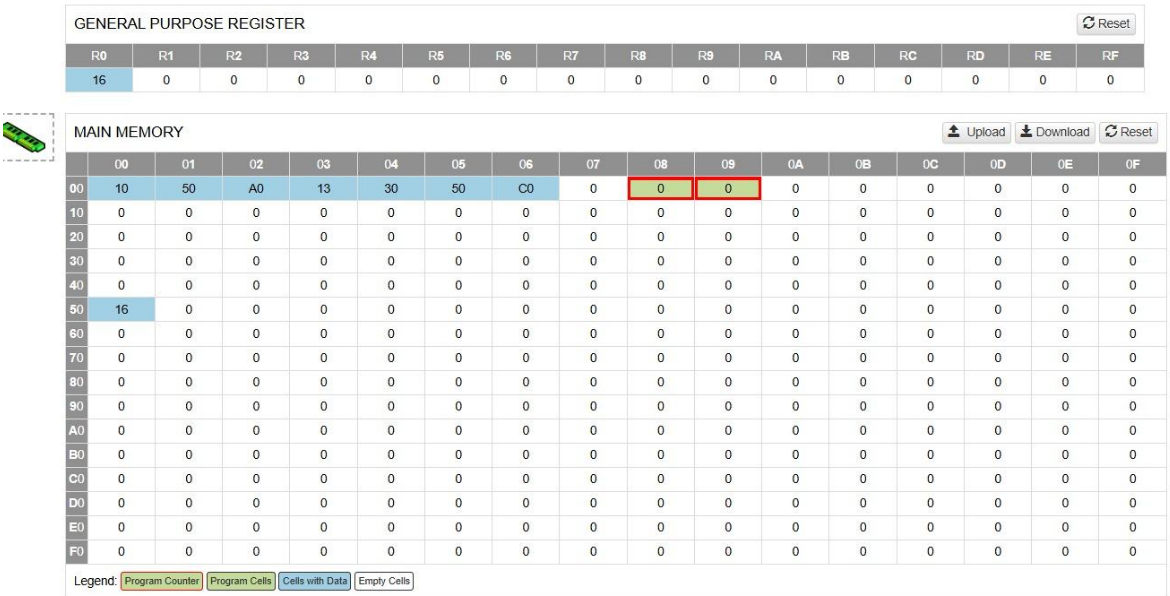
MAIN MEMORY

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
00	11	83	12	84	2F	FF	96	F2	20	01	56	06	53	16	20	00
10	27	80	88	37	B8	1A	32	85	C0	00	31	85	C0	00	00	00
20	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
30	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
40	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
50	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
60	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
70	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
80	00	00	00	09	07	09	00	00	00	00	00	00	00	00	00	00
90	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
B0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
C0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
D0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
F0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

Legend: Program Counter Program Cells Cells with Data Empty Cells

Explanation for the program:
The program begins with two LOAD commands: moves datas from their memory addresses to Registers R1 and R2. The next part of the program prepares for subtraction by changing the sign of the second number with a LOAD command putting FF into RF, and then a XOR operation is done between RF and R2 with the result going into R6. Next, the LOAD command puts the value 1 into R0, this 1 is then added to the content of R6 using the ADD instruction to finalize the two's complement. The main calculation follows with the ADD instruction again, which adds the original number in R1 to the negative value in R6, and the result goes to R3. Finally, the program loads the value 0 into R0 and the value 80 into R7. Then the AND command checks the sign bit of R3 and places the result in R8. The JUMP command executes and the STORE instruction ultimately saves the greatest number to the Memory address. And lastly the HALT command ends the entire program execution.

Task 3



GENERAL PURPOSE REGISTER

R0	R1	R2	R3	R4	R5	R6	R7	R8	R9	RA	RB	RC	RD	RE	RF
16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

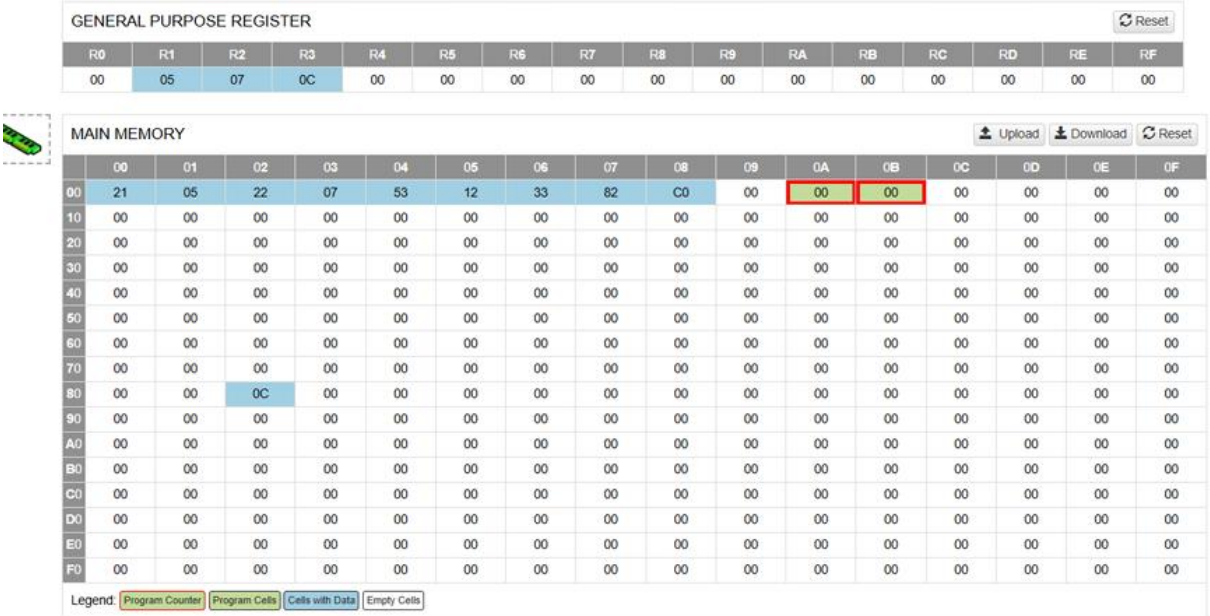
MAIN MEMORY

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
00	10	50	A0	13	30	50	C0	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
40	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
50	16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
60	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
70	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
80	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
90	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
A0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
B0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
D0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
F0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Legend: Program Counter Program Cells Cells with Data Empty Cells

Explanation for the program:
The program starts with the LOAD command, which copies the data in the memory into the Register R0. Next, the ROTATE instruction shifts the data in R0 right by 3 bits, this process occurs entirely within the register. Following the rotation, the STORE command moves the final modified byte from Register R0 and writes it back to Memory address. Finally, the HALT command is executed and it stops the program.

Optional task:



GENERAL PURPOSE REGISTER

R0	R1	R2	R3	R4	R5	R6	R7	R8	R9	RA	RB	RC	RD	RE	RF
00	05	07	0C	00	00	00	00	00	00	00	00	00	00	00	00

MAIN MEMORY

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
00	21	05	22	07	53	12	33	82	C0	00	00	00	00	00	00	00
10	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
20	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
30	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
40	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
50	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
60	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
70	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
80	00	00	0C	00	00	00	00	00	00	00	00	00	00	00	00	00
90	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
B0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
C0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
D0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
F0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

Legend: Program Counter, Program Cells, Cells with Data, Empty Cells

Explanation for the program:

The original program used Opcode 1 to load data from memory addresses to registers. But this one uses Opcode 2 to place the constants directly into the registers. This is faster because it eliminates the two memory access steps required by the original program.